

DGX Spark Dual Cooler

Screwless 3D-printed cooling cage for two DGX Spark / ASUS Ascent GX10

ASSEMBLY GUIDE · FUSION MODEL v33 · SEPTEMBER 2026

3 x 120 mm

fans

ZERO

screws / inserts

206 x 253 x 190

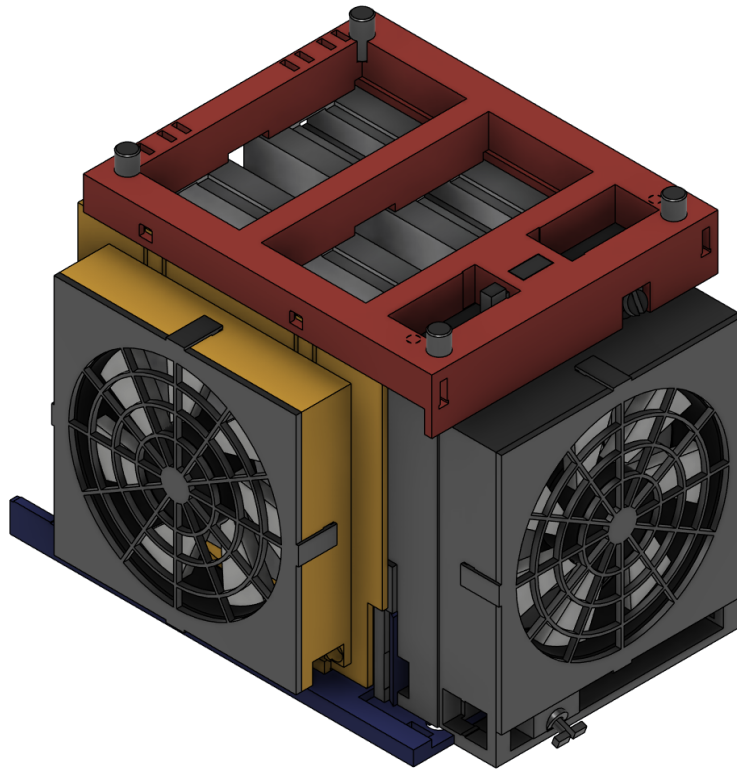
mm assembled

~580 g

PETG @ 20% infill

15

printed parts



COLOUR KEY - used throughout this guide

■ Base tray

■ Side fan arms

■ Front plenum

■ Top brace

■ Fans & grilles

Everything clips, slides or wedges together. There is not a single screw, heat-set insert or drop of glue in the assembly.

Read pages 2 and 3 before you start printing - the fan you buy decides which grille you print, and two of the ten parts need support material.

What you print

Part	Qty	Size mm	Print orientation	Support	cm3
PRT_01 Base tray	1	176 x 220 x 46	floor down	none	205
PRT_02 Front plenum	1	139 x 180 x 66	back plate down	yes	284
PRT_03 Side fan arm	2	172 x 166 x 40	flat, collar up	yes	122 ea
PRT_04 Top brace	1	156 x 190 x 30	top face down	none	240
PRT_05 Fan grille (26 mm fans)	3	135 x 135 x 21	grille face down	none	29 ea
PRT_05B Fan grille (25 mm fans)	3	135 x 135 x 21	grille face down	none	29 ea
PRT_06 Wedge key	1	10 x 6 x 41	on edge	none	1.4
PRT_07 Stacking peg	4	dia 10 x 20	standing	none	1.6 ea
PRT_08 Button pin, long	1	15 x 31 x 72	widest face down	none	1.2
PRT_09 Button pin, short	1	10 x 10 x 45	oblique	none	0.6

Every STL and the 3MF is already in its correct orientation, sitting on Z = 0. Drop them in and slice - do not rotate. Print one grille variant only: PRT_05B has 2.2 mm clamp pads for 25 mm fans, PRT_05 has 1.2 mm pads for 26 mm fans.

TWO PARTS NEED SUPPORT

The side fan arm and the front plenum each carry a large flat panel that starts in mid-air, and neither has an orientation that avoids it. Tree supports under the arm plate are enough; the plenum wants normal supports under its front face. The other eight parts print with nothing, worst unsupported reach 5 mm.

PRINT SETTINGS

- PETG. Every latch, hook and pin leg is a printed spring.
- 0.4 mm nozzle, 0.2 mm layers.
- 4 perimeters minimum - thin walls halve every snap feature.
- 20 % infill. Nothing is structural beyond the flexures.
- No brim needed on any part.



Collar top wall, notched at both ends for the unit feet

PLATE GROUPING - HOW THE 3MF IS LAID OUT

Plate	Contents
1	Base tray
2	Front plenum
3 and 4	Side fan arm, one per plate
5	Top brace + wedge key + 4 stacking pegs + both button pins
6 to 8	Fan grille x 3, 25 mm variant (PRT_05B)
9 to 11	Fan grille x 3, 26 mm variant (PRT_05) - print one set or the other

Standard 3MF has no plate concept, so most slicers open all objects at once in a grid. Use "arrange all" or assign plates by hand from this table. The grilles are 135 mm square, so only one fits a 248 x 250 mm bed at a time.

What you buy

FANS - 3 NEEDED

Thermalright TL-C12C-S X3

120 x 120 x 25 mm, 3-pack, 4-pin PWM, S-FDB bearing. 1550 rpm, 66.2 CFM, 1.53 mmH₂O, 25.6 dBA, 0.20 A each.

Amazon ASIN B0BN19KSW2

Buy the plain version, not the ARGB one. Print grille PRT_05B (2.2 mm pads) for these. Any standard 120 x 120 fan with a 105 mm hole pitch fits - print PRT_05 instead if yours is 26 mm thick.

POWER & SPEED CONTROL - 1 NEEDED

Wathai 36 W fan speed controller

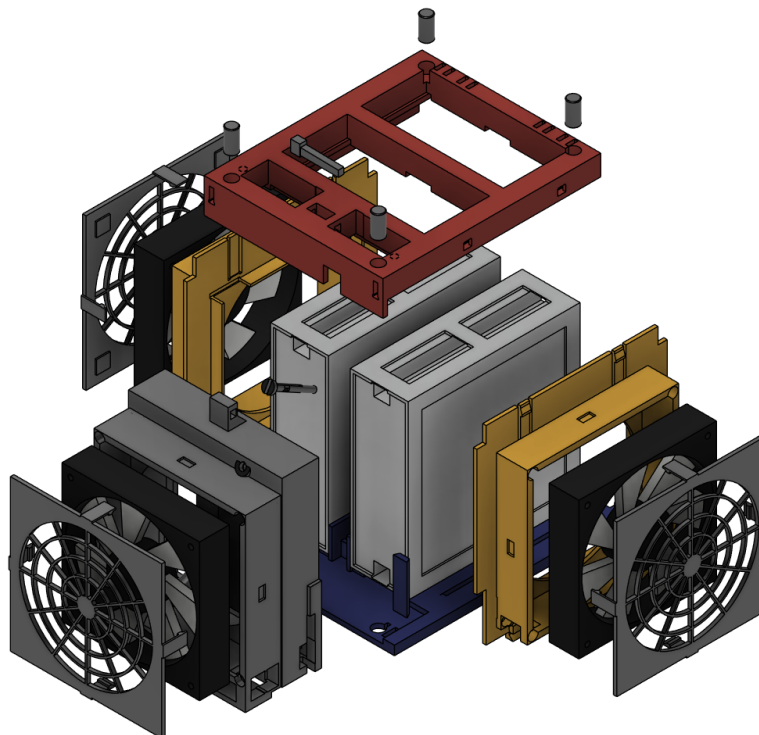
110-240 V AC to 4-12 V DC, 3 A, with a 4-port 3-pin/4-pin splitter. The knob sets voltage; all the heat stays in the external brick.

Amazon ASIN B0F4KG6MM1

One brick runs all three fans - they draw about 0.6 A of the 3 A available. The fourth port is spare for a second cage.

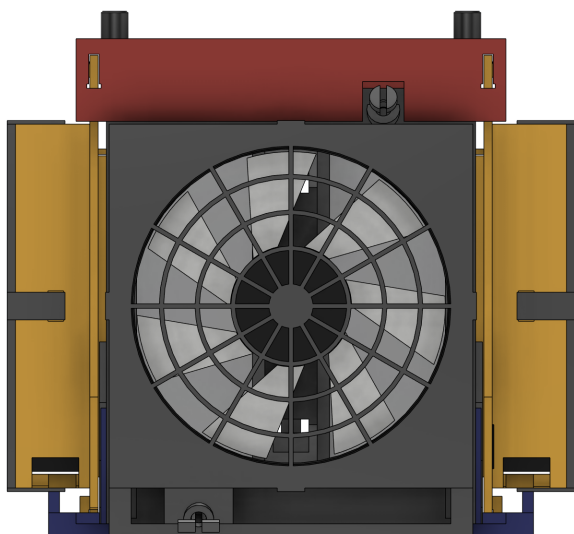
ALSO USEFUL

- About eight 2.5 x 100 mm zip ties for the cable channels.
- A small flat screwdriver, 6 mm or narrower, for releasing the arm latches at teardown.
- A QSFP DAC stacking cable of your choice - allow 95 mm of clearance behind the cage for its bend radius, 120 mm is comfortable.

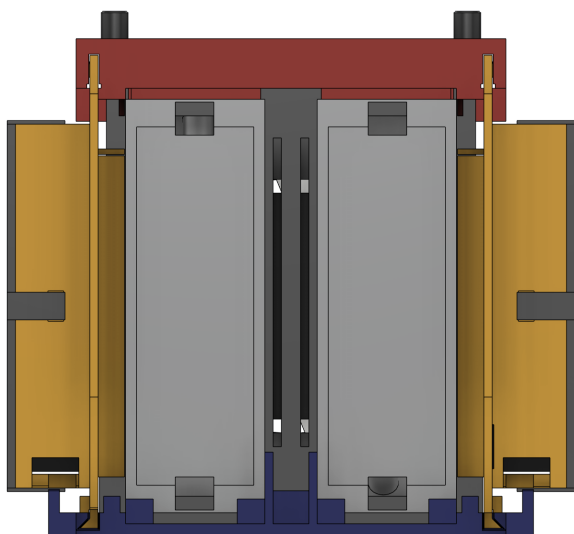


How the ten unique parts stack up

How it holds together



FRONT - intake, front fan, both button pins



REAR - ports, QSFP cable, exhaust, glands

The two units are **MIRRORED**. One is flipped end-for-end so that both filtered bottom panels face outward into the side fans. A consequence is that one unit's power button sits near the top of the front face and the other near the bottom - which is why there are two different button pins.

Air path: each side fan blows inward through the unit's filtered bottom panel; the front fan pressurises a sealed plenum that feeds both front intakes, plus a centre duct that jets air down the 19 mm gap between the units. Everything exhausts rearward, and the rear is left deliberately clear.

THE FIVE THINGS THAT HOLD IT TOGETHER

Joint	Mechanism	Numbers
Arm to tray	45 deg half-dovetail, slides in from the rear	0.12 mm clearance, 150 mm travel
Arm to brace	Two cantilever latch barbs into catch pockets	4.6 N to release, 0.90 mm engagement
Grille to collar	Four cantilever hooks over four notches	2.8 N per hook, ~11 N to click on
Plenum to brace	Tusk tenon through a mortise, wedge key	0.65 mm of taper to tighten
Button pin to tube	Split shaft, two spring legs, snap barb	0.45 mm squeeze of 1.20 available

BEFORE YOU START

Work on a clear bench with the tray front edge toward you, and keep the two arms identified - they are the same printed part used twice, the second one rotated 180 degrees. Nothing in this build needs force beyond a firm push; if a part will not go, something is the wrong way round.

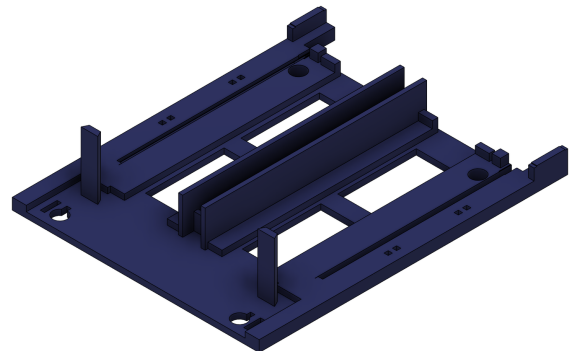
Assembly

1 Set down the base tray

Put the tray on the desk with the front edge - the end carrying the two dovetail posts - toward you. Every other part locates off this one, so get it square and leave clearance behind it.

The rear end is the one with the wide side ears and the two cable glands.

Nothing is fixed to the desk. The cage simply sits on the tray floor.

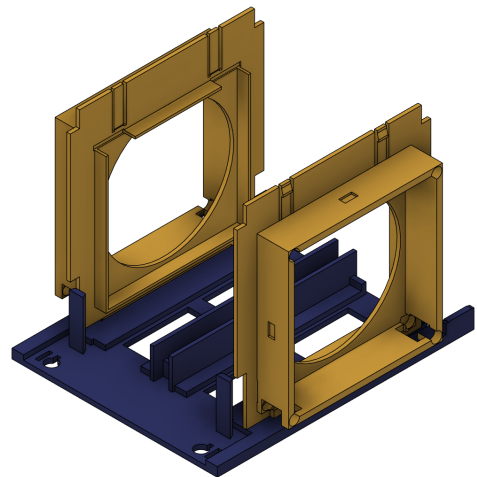


2 Slide in the two side fan arms

Both arms are the same printed part, the second one rotated 180 degrees. Start each arm at the REAR of the tray with its dovetail tongue in the slot, then slide it forward about 150 mm until it hits the hard stop.

The clearance is 0.12 mm per flank - it should slide with light thumb pressure and no rattle.

Keep the rear ears clear: anything on them taller than 17 mm blocks the arms.



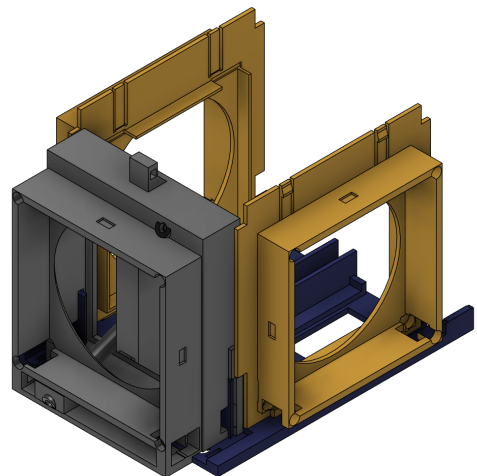
Assembly

3 Drop the front plenum on

Lower the plenum straight down over the two dovetail posts at the front of the tray. It engages only one way round. The last 40 mm should drop under its own weight.

The dovetails lock it fore-aft and side-to-side; the brace will stop it lifting.

The two round tube ends visible inside are the power-button pin guides.

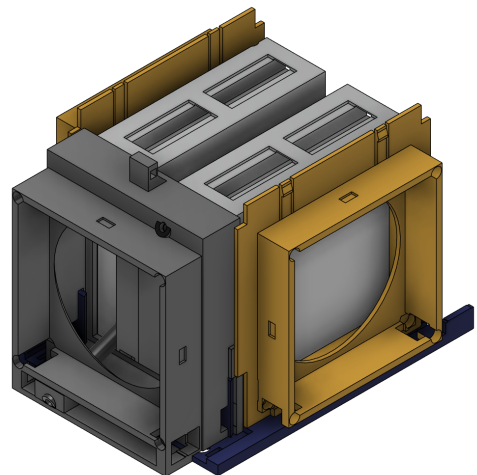


4 Lower in the two units

Stand each unit on edge and lower it into its cradle. The second unit goes in flipped end-for-end, so that both filtered bottom panels face outward toward the side fans.

Connect the QSFP stacking cable now - it is far easier before the brace goes on.

The units are held inboard by the tray centre wall and outboard by the arm collar, leaving 0.40 mm of play.



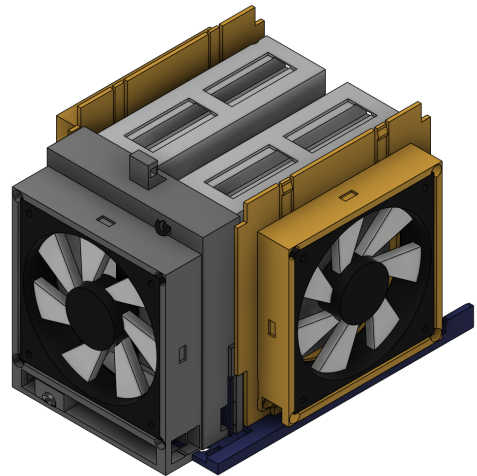
Assembly

5 Push in the three fans

Each fan pushes into its collar from the outside and seats on the round sealing land at the bottom of the pocket. Check the arrow on the fan frame: both side fans must blow INWARD, and the front fan must blow toward the rear.

Drop the lead into whichever corner channel it lands nearest - all four corners are relieved, so any rotation works.

The fan sits about 1 mm proud of the collar face. The grille takes up that gap.

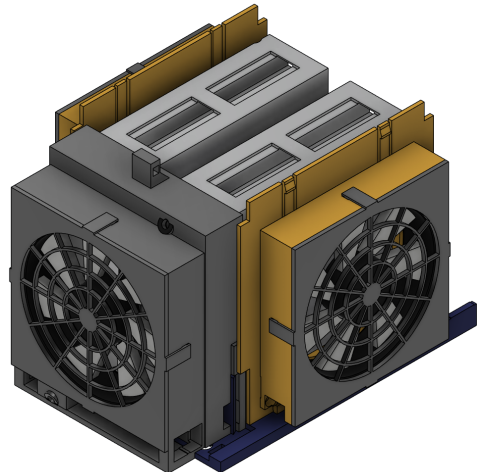


6 Clip on the three grilles

Line each grille up so its four hooks sit over the four notches, then press evenly until all four click - about 11 N. Check the variant matches your fan thickness before you push.

PRT_05B for 25 mm fans, PRT_05 for 26 mm. The wrong one leaves the fan loose and buzzing.

To swap a fan later, pry off that grille only. Nothing else comes apart.



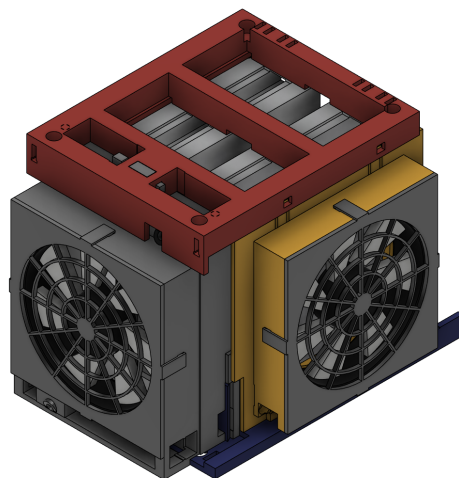
Assembly

7 Fit the top brace, then the wedge key

Lower the brace straight down. The tops of both arms enter the grooves in its underside, and about 5 mm from home the two latch barbs cam over and snap into their catches. Then slide the wedge key in from the left, through the bridge and the plenum tenon, and tap it home.

The key is the only thing resembling a fastener in the build. It locks the brace to the plenum.

It has 0.65 mm of taper, so tap it a little further if the joint ever feels slack.

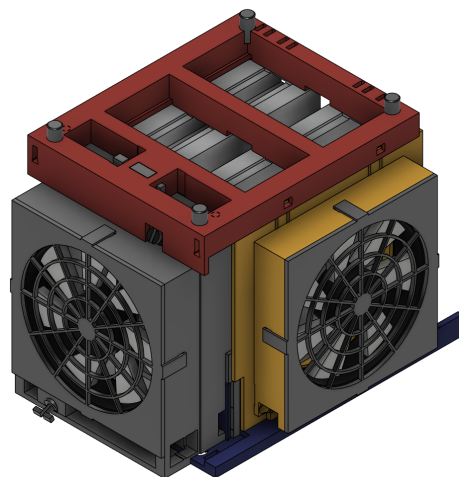


8 Pegs and button pins

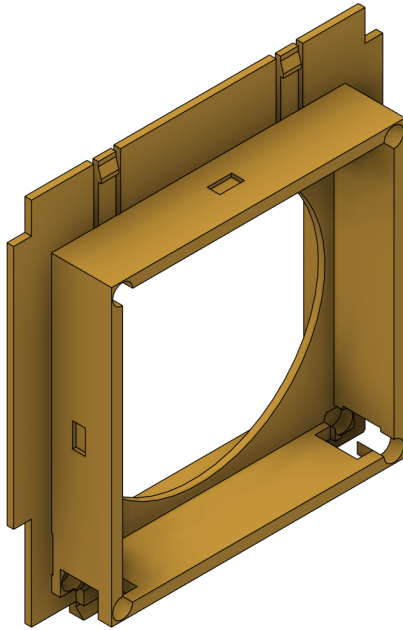
Press the four stacking pegs into the top of the brace, then snap each button pin into its guide tube from the front. The long pin serves the unit whose button is high, the short one the unit whose button is low.

Each pin needs about 0.45 mm of leg squeeze to pass the mouth, and is captive once in.

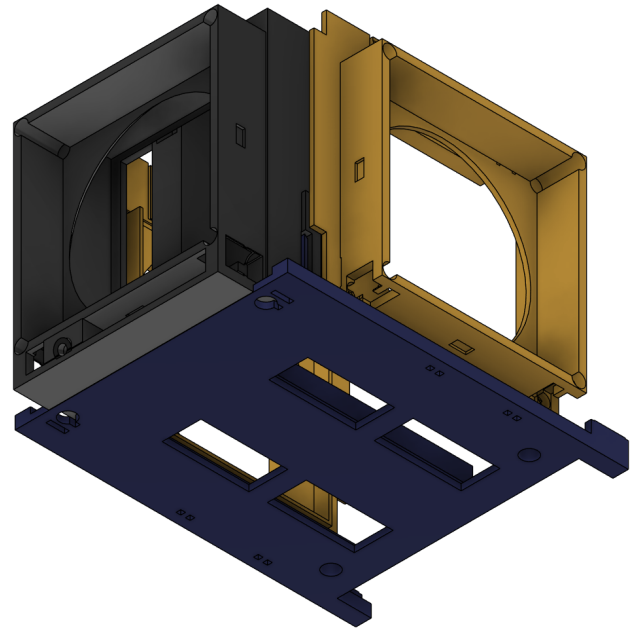
To release an arm latch at teardown, push the flat pad square through the window in the brace with a 6 mm blade - about 4.6 N.



Routing the fan leads



COLLAR - relieved corners and the floor exit



UNDERSIDE - floor exits and the run to the rear

All three fan leads run inside the cage and come out together at the rear, where the brick and splitter sit. Nothing crosses the rear ports and nothing sits in the hot exhaust - the run is entirely outboard of the exhaust footprint and below the port strip.

- Side fans: the lead drops down whichever pocket corner it sits in, through the floor exit, into the tunnel between the tray ear and the underside of the arm collar.
- Front fan: the lead leaves through a side window in the plenum skirt and joins the same tunnel.
- Both then run rearward through the gland and out of the back of the tray. Zip-tie the bundle at the gland so nothing can creep back into a fan.

Every restriction on the route clears a 14 x 8 mm fan connector: arm cable windows 16.5 x 10.0 mm, plenum skirt windows 21.5 x 10.0 mm per side, plenum rear slot 10.5 x 16.5 mm per side, fan-pocket floor slots 15.5 mm on the arm and 14.3 mm on the plenum.

Power, first run and service

WIRING

Plug all three fans into the 4-port splitter. The fans are 4-pin PWM but the controller sets speed by voltage - the PWM line simply floats, which is fine.

Splitter into the 4-12 V brick, brick into the wall. Keep the brick outside the cage; all of its heat stays in the brick, which is the whole reason for choosing it.

SETTING THE SPEED

Start the knob at minimum and wind it up. Below roughly 5 V the fans may not spin up at all - that is normal for voltage control, not a fault.

Three fans draw about 0.6 A of the available 3 A, so the fourth port is free for a second cage.

SERVICE

- Swapping a fan: pry off that grille only, everything else stays together.
- Full teardown: drive the wedge key out, release both arm latches through the brace windows, lift the brace, then reverse the build.
- The arms slide out to the rear. Do not lift a unit out before the brace is off - the capture ribs straddle its top edge.

IF YOU HAVE ARMS FROM AN EARLIER RELEASE

The GX10 has four rubber feet on its bottom panel, and that panel is the face looking out into the side fan, so the feet land on the shroud collar. Arms printed from the current files are already notched to clear them. On older arms, cut the collar top wall away from each corner to the far edge of the latch snap, both sides, both ends, both arms.

WHAT IS NOT PROVEN YET

- Nothing since the first build has been printed. Every figure in this guide is measured from the solid model, not from a part in hand.
- The centre duct is a calculated design. Its two exit slots are tapeable, so an A/B test is easy - tape them, run the same load, compare.
- Latch, hook and pin-leg stiffness are calculated, not measured.

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<https://github.com/koldfrontier/dgx-spark-ops>

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